

txt2tex SEM/SBM Reference

Software Engineering Mathematics & State-Based Modelling: logic, sets, sequences, proofs, Z schemas, operations, Δ/Ξ , schema calculus.

<https://github.com/jmf-pobox/txt2tex>

Document Structure

=== Title ===	<code>\section *{Title}</code>
* Solution N **	Bold heading with spacing
(a) Text	Part label
TEXT: paragraph	Prose block (smart formula detection)
LATEX: <code>\cmd</code>	Raw L ^A T _E X passthrough
PAGEBREAK:	Page break
LINEBREAK:	Vertical gap (<code>\medskip</code>) — between blocks, no page break
TITLE: ...	Document title
AUTHOR: ...	Author name
DATE: ...	Date string
BIBLIOGRAPHY: f.bib	BibTeX file
[cite key]	<code>\citep {key}</code>

Logic

<code>p land q</code>	$p \wedge q$
<code>p lor q</code>	$p \vee q$
<code>lnot p</code>	$\neg p$
<code>p => q</code>	$p \Rightarrow q$
<code>p <=> q</code>	$p \Leftrightarrow q$
<code>forall x : S P</code>	$\forall x : S \bullet P$
<code>exists x : S P</code>	$\exists x : S \bullet P$
<code>exists_1 x : S P</code>	$\exists_1 x : S \bullet P$
<code>lambda x : S E</code>	$\lambda x : S \bullet E$
<code>mu x : S P</code>	$\mu x : S \bullet P$

Note: Use `land`, `lor`, `lnot`. English `and/or/not` are not supported.

Sets & Types

<code>x elem S</code>	$x \in S$
<code>x notin S</code>	$x \notin S$
<code>A subset B</code>	$A \subseteq B$
<code>A union B</code>	$A \cup B$
<code>A inter B</code>	$A \cap B$
<code>A setminus B</code>	$A \setminus B$
<code>power S</code>	$\mathcal{P} S$
<code>F S</code>	$\mathcal{F} S$
<code>{x : S P}</code>	Set comprehension
<code>A cross B</code>	$A \times B$
<code>N / Z / N1</code>	$\mathbb{N} / \mathbb{Z} / \mathbb{N}_1$
<code>n..m</code>	$n .. m$

Z Notation Blocks

<code>given A, B</code>	Given types: $[A, B]$
<code>T ::= a b</code>	Free type
<code>name == expr</code>	Abbreviation
<code>axdef / schema</code>	Name / gendef [X]
<code>declarations</code>	Variable declarations
<code>where</code>	Separator
<code>predicates</code>	Constraining predicates
<code>end</code>	Close the block

Usage

<code>txt2tex f.txt</code>	$\rightarrow$ f.tex + f.pdf
<code>txt2tex f.txt --tex-only</code>	$\rightarrow$ f.tex only
<code>txt2tex -i</code>	Interactive REPL
<code>txt2tex --check-env</code>	Verify dependencies

Relations

<code>A <-> B</code>	$A \leftrightarrow B$
<code>x -> y</code>	$(x \mapsto y)$
<code>dom R / ran R</code>	$\text{dom } R / \text{ran } R$
<code>A dres R</code>	$A \triangleleft R$
<code>A dsub R</code>	$A \triangleleft R$
<code>R rres B</code>	$R \triangleright B$
<code>R rsub B</code>	$R \triangleright B$
<code>R oplus S</code>	$R \oplus S$ (override)
<code>R o9 S</code>	$R \circ S$ (forward comp.)
<code>R comp S</code>	$R \circ S$ (backward comp.)

Functions

<code>A pfun B</code>	$A \rightarrow B$
<code>A fun B</code>	$A \rightarrow B$ (total)
<code>A pinj B</code>	$A \rightarrowtail B$
<code>A inj B</code>	$A \rightarrowtail B$
<code>A surj B</code>	$A \twoheadrightarrow B$
<code>A bij B</code>	$A \xrightarrow{\sim} B$
<code>A ffun B</code>	$A \dashrightarrow B$ (finite)
<code>f(x)</code>	Function application

Sequences

<code><> / <a, b></code>	$\langle \rangle / \langle a, b \rangle$
<code>s ^ t</code>	$s \frown t$ (concatenation)
<code>seq S / seq1 S</code>	$\text{seq } S / \text{seq}_1 S$
<code># s</code>	$\#s$ (length)

Truth Table Prop. Logic

You write:

TRUTH TABLE:
p | q | p => q
T | T | T
T | F | F

F | T | T
F | F | T

You get:

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

txt2tex Proofs — By Example

Each example shows what you type and what txt2tex renders.

Proof Tree Syntax

indent	Premises indented under conclusion
[rule]	Justification label
[rule [n]]	Discharge assumption n
[n] expr	Boxed assumption labelled n
::	Sibling premises

Rules: land intro, land elim 1/2, lor intro 1/2, lor elim, => intro/elim, <=> intro/elim, lnot intro/elim, forall intro/elim, exists intro/elim.

How Proof Trees Work

In txt2tex, proof trees are written **bottom-up**: the conclusion is the first line, and its premises are indented below it. Think of it as reading the inference rule from conclusion to justification:

conclusion	First line (least indented)
premise	Indented = supports the line above
[rule]	Justification in brackets
[n] expr	Boxed assumption (labelled n)
[rule [n]]	Discharge assumption n
::	Marks sibling premises

Deeper indentation = deeper in the tree. Two premises at the same indent level are siblings that jointly support their parent.

Equivalence Chain Prop. Logic

You write:

```
EQUIV:
lnot (p land q)
lnot p lor lnot q  [De Morgan]
```

You get:

$$\neg (p \wedge q) \\ \Leftrightarrow \neg p \vee \neg q \quad [\text{De Morgan}]$$

EQUIV: joins steps with $\Leftrightarrow$. Use for *predicate* equivalences.

Modus Ponens Proof Trees

You write:

```
PROOF:
q [=> elim]
p => q
p
```

You get:

$$\frac{p \Rightarrow q \quad p}{q} \Rightarrow \text{-elim}$$

$\wedge$ -intro Proof Trees

You write:

```
PROOF:
p land q [land intro]
p
q
```

You get:

$$\frac{p \quad q}{p \wedge q} \wedge \text{-intro}$$

$\Rightarrow$ -intro with discharge Proof Trees

You write:

```
PROOF:
p => p [=> intro [1]]
[1] p
```

You get:

$$\frac{\neg p \neg^{[1]}}{p \Rightarrow p} \Rightarrow \text{-intro}^{[1]}$$

[1] p = boxed assumption. [=> intro [1]] discharges it.

$\vee$ -elim / case analysis Proof Trees

You write:

```
PROOF:
r [lor elim]
p lor q
:: p => r
[1] p
r [from p]
:: q => r
[2] q
r [from q]
```

You get:

$$\frac{p \vee q \quad \frac{\neg p \neg^{[1]}}{r} \quad \frac{\neg q \neg^{[2]}}{r}}{r} \vee \text{-elim}$$

:: = sibling premises (case arms).

Equality Chain Induction

You write:

```
EQUAL:
0 + 0
0      [0+0 = 0]
length(<>)
[length(<>) = 0]
```

You get:

$$0 + 0 \\ = 0 \quad [0+0 = 0] \\ = \text{length}(\langle \rangle) \quad [\text{length}(\langle \rangle) = 0]$$

EQUAL: = expression chains (=). Use for $\mathbb{N}$ /sets/sequences.

txt2tex Schemas — By Example

Schema definition, where-clause patterns, inclusion, calculus.

Schema Definition

A named schema introduces a signature (declarations) and an invariant (where clause). The block opens with **schema** *Name* and closes with **end**.

You write:

```
schema Counter
  size : N
  nums : seq N
where
  size = # nums
end
```

You get:

<i>Counter</i>	
<i>size</i> : $\mathbb{N}$	
<i>nums</i> : seq $\mathbb{N}$	
<i>size</i> = # <i>nums</i>	

axdef introduces global constants instead of a named signature;
gendef [X] parameterises the block by a type.

Where-Clause Patterns

Three common constraint shapes; one schema may mix them.

1. Simple comparison.

You write:

```
schema Bounded
  size : N
where
```

```
  size < 10
end
```

2. Set comprehension.

You write:

```
schema Inventory
  items : seq N
  stock : N
where
  stock = # {x : ran items | x > 0}
end
```

“the count of strictly positive items in items.”

3. Quantifier.

You write:

```
schema NonNeg
  s : seq Z
where
  forall i : 1..#s | s(i) >= 0
end
```

“every element of *s* is non-negative.”

Schema Inclusion

Pull another schema’s signature into the current one. Used to build operation schemas from state schemas.

SName	bare inclusion (signature + invariant)
Delta Counter	$\Delta Counter$ (before-/after-state pair)
Xi Card	$\Xi Card$ (state read-only)
theta S	θS (binding of current state)
theta S’	$\theta S'$ (binding of after-state)

Schema Calculus

Combine schemas to form new ones. The horizontal form **defs** names the result.

Name **defs** RHS — horizontal definition:

Name **defs** Schema *Name* $\hat{=}$ *Schema*

N **defs** Delta C *N* $\hat{=}$ ΔC

N **defs** [d : T | P] inline schema text

N[X] **defs** G[X] generic form

defs RHS operators:

S[old/new] rename component

S[a/b, c/d] multiple renames

S ; T *S* ; *T* (composition)

S » T *S* $\gg$ *T* (piping)

S hide (x, y) *S* \ (*x*, *y*)

S project T *S* $\upharpoonright$ *T*

Identifier Decoration

count’ *count'* (after-state)

in? *in?* (input)

out! *out!* (output)

x?’ *x?’* (runs may combine)

’sunk’ ‘sunk’ (string literal)

Decoration attaches to any identifier. Reserved words (**theta**, **defs**, **hide**, **project**, **Delta**, **Xi**) cannot be decorated.